

Agent Harness Awareness: Turning Verified Telemetry into Operational Control for Tool-Using Agents

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Abstract

Autonomous agents increasingly act through tools, yet their harnesses keep the operational facts that determine whether those actions will succeed—latency, failures, route health, deadlines, quotas, and resource pressure—outside the agent’s decision loop. This creates **operational blindness**: semantically equivalent actions can have radically different operational consequences while appearing identical to the model. We introduce **Agent Harness Awareness**, a trust-separated control architecture that converts bounded, fresh, host-verified telemetry into an agent decision input while keeping measurement, enforcement, and evaluation outside the model. We evaluate the tools-and-network instance in two separately analysed frozen studies. In the primary 216-decision direct-provider ToolRoute ablation across GPT-5.6 Sol, Claude Sonnet 5, and Gemini 3.7 Flash, verified tool-health state reduced mean observable policy regret by **96.9%** against a neutral same-shape interface (4,252 ms to 131 ms) and raised completion from **69.4% to 95.8%**. In a separate 27-decision end-to-end replication using live local TCP/HTTP execution, OpenTelemetry client spans, and Toxiproxy-injected faults, verified state selected the observable-best route and completed the live action in **9/9** model-by-fault cases, versus **5/9** completion for the neutral control. These results establish a practical systems principle: telemetry should not end at human dashboards. Selected, provenance-preserving operational state can serve as a high-leverage control input for autonomous agents.

1. From operational blindness to closed-loop control

An agent can understand a user request and every tool’s semantics yet still make an avoidable operational error. The missing information is often not semantic; it is the current state of the harness: which route is slow, failing, rate-limited, stale, resource-constrained, or no longer economical. When the harness knows this state but the agent does not, equivalent APIs remain indistinguishable at the moment the agent selects one.

This is a general agent-runtime problem, not merely a routing problem. Harnesses already observe four high-value operational dimensions: **tools and network** (route health and circuit state); **runtime** (deadlines, queueing, and concurrency); **hardware and sandbox** (memory headroom, CPU pressure, storage); and **economics** (context headroom, token burn, quota, and cost). A harness-aware agent can use these facts for alternate-route selection, deadline-aware decomposition, checkpointing before budget exhaustion, resource-aware concurrency, and backpressure-aware coordination.

We call this gap **operational blindness**. Agent Harness Awareness closes it by reducing bounded, freshness-labelled, host-verified execution state into an agent decision interface. The host retains authority over collection, reduction, safety, and enforcement. The model may adapt its next strategic action, but cannot rewrite measurements, disable constraints, or modify the evaluator.

This paper contributes a general systems paradigm, a trust-separated state interface, a frozen 216-decision multi-model information ablation, and a separately analysed 27-decision OpenTelemetry/Toxiproxy transport replication. It is evidence that what an agent knows about its harness can be as consequential as what it knows about the user task.

2. Agent Harness Awareness

At decision time t , let x_t denote the task and tool interface, h_t the host state, and $z_t = \varphi(h_t)$ a bounded host-generated projection. The proposed **Substrate State Telemetry (SST) contract** covers Tools (success history, latency, error/circuit state), Runtime (deadlines and exit state), Hardware (memory, CPU, disk, processes), and Economics (context headroom, quota, rate, cost). ToolRoute evaluates the Tools dimension; the other dimensions are direct future instantiations, not pooled evidence.

Harness Awareness complements rather than replaces deterministic controls. The data plane—kernel, proxy, circuit breaker, retry policy, and scheduler—continues to own immediate recovery and safety. The agent control plane acts when current state changes strategy: choose a route, defer work, adopt a degraded-mode plan, or allocate time and budget differently.

3. Two complementary ToolRoute studies

ToolRoute evaluates the tools-and-network dimension through two deliberately distinct studies. Study 1 is the primary controlled information ablation; Study 2 is an independently reported systems replication. Their data are never pooled.

Study	Question	Evidence	Role
Study 1	Does truthful state improve route selection when the interface is held constant?	216 frozen provider decisions; seeded monitor	Primary behavioral estimate
Study 2	Does the effect survive an OTel-instrumented live HTTP tool path?	27 frozen decisions; OTel, Toxiproxy, live action	End-to-end replication

3.1 Study 1: controlled information ablation

The agent retrieves a read-only customer record through one of two functionally equivalent routes, `tool_alpha` and `tool_beta`, or chooses `wait`. The host owns monitoring, schedules, execution, and an external observable-cost oracle. Each decision is an independent full-checkpoint episode; models have no shell, filesystem, network-tool, schedule, oracle, or artifact access.

Condition	Model-visible information
A: task only	Task and equivalent tool descriptions; no operational projection.
B: neutral envelope	Same telemetry-shaped envelope, fields, row order, and tool rows as C, with equal neutral values.
C: verified state	Same envelope with fresh host-owned facts for current route state.

A is the deployment baseline: task and tool semantics alone, approximating conventional agent behavior without harness state. **B is the structural control:** the same telemetry-shaped interface with all route-relevant values neutralized. **C is the intervention:** verified current state. Thus A-to-C measures practical deployment gain, while B-to-C isolates the informational contribution of telemetry from interface shape and formatting. B is not an exact provider-token-matching control; input-token counts are descriptive.

We report two complementary outcomes. **Observable policy regret** is the avoidable observable operational cost of the selected action: $r_{\text{obs}}(a,z) = c_{\text{obs}}(a,z) - \min_{a'} c_{\text{obs}}(a',z)$. The host computes c_{obs} using only canonical monitor facts rendered in C, combining monitored latency with predeclared failure, circuit-open, and missed-record penalties. It is reported in milliseconds; **0 ms regret means that the agent selected an observable-best action with no excess operational cost, not that the request itself took zero time.** The evaluator never uses hidden latent anomalies unavailable to the model. **Completion** is a separate binary outcome: whether the selected route successfully retrieves the requested record; in Study 2, whether the post-decision live HTTP action succeeds. A route can complete yet still have positive regret when it was avoidably slower or riskier.

The frozen direct-provider cohort contains 216 independent decisions (three model families $\times$ six seeds $\times$ four turns $\times$ three conditions); every attempt retains its prompt, sanitized request/response, snapshot, tool event, evaluation result, and finalization hashes. No request was retried.

3.2 Study 2: live OpenTelemetry transport replication

The separately labelled v0.2 replication preserves the A/B/C information intervention but moves to a live instrumented tool path. Each episode makes three monitor requests per route through local TCP/HTTP proxies. Standard OpenTelemetry HTTP client spans provide duration and outcome facts; a host reducer converts them to the decision snapshot. Toxiproxy injects a predeclared latency, connection, or HTTP failure on one route, and the selected route is then executed through the same still-live proxy. The frozen replication contains 27 independent decisions (three models $\times$ three fault regimes $\times$ A/B/C) and retains raw spans, reducer events, provider records, action spans, results, and hashes.

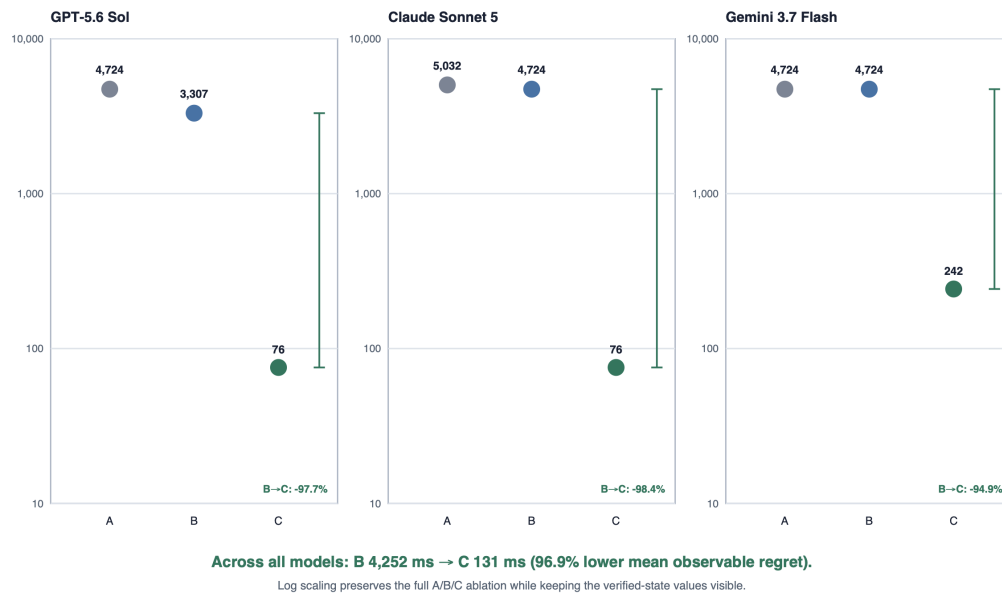
4. Results

4.1 Study 1: controlled information ablation

Against the task-only deployment baseline A, verified state C reduces mean observable policy regret from 4,826.45 ms to 131.12 ms: a **97.3% reduction**. Completion rises from 47/72 (65.3%) to 69/72 (95.8%), a **30.6-point increase**. The structural B-to-C comparison reaches the same conclusion while holding interface shape constant: regret falls from 4,251.54 ms to 131.12 ms (**96.9% lower**) and completion rises from 50/72 (69.4%; Wilson 95% CI 58.0-79.0%) to 69/72 (95.8%; Wilson 95% CI 88.3-98.5%), a **26.4-point increase**. Together, these comparisons show both practical deployment value and the informational contribution of verified telemetry.

Figure 1. Verified state collapses observable regret across all information conditions

Condition means on a logarithmic scale; n = 24 independent direct-provider decisions per point. Lower is better.



Frozen ToolRoute v1.0 cohort. Condition means are independent samples; points are not paired trajectories.

Figure 1. Mean observable policy regret for all three information conditions in the frozen 216-decision cohort. Log scaling keeps verified-state values visible; A, B, and C are independent condition means, not trajectories.

Condition	Role	Mean regret	Completion
A	Task-only deployment baseline	4,826.45 ms	47/72 (65.3%)
B	Neutral structural control	4,251.54 ms	50/72 (69.4%)
C	Verified host state	131.12 ms	69/72 (95.8%)

4.2 Study 2: live transport replication

The transport replication reproduces the central result without relying on the synthetic monitor. The task-only deployment baseline A completed 6/9 live actions (66.7%) with 6,734.30 ms mean observable policy regret. The neutral structural control B completed 5/9 actions (55.6%) with 8,956.37 ms regret. Verified state C selected the observable-best route in **9/9 cases**, completed **9/9 live actions**, and incurred **0.00 ms** regret. Thus C improves both the practical task-only baseline and the same-shape structural control, while the B-to-C comparison isolates the information effect.

Figure 2. Verified telemetry improves every live transport decision

Frozen OpenTelemetry/Toxiproxy v0.2 replication: 27 independent remote-model decisions; live selected-route execution after each decision.



Figure 2. Study 2 live transport replication. The upper panel shows that all nine verified-state C cases selected the observable-best route and completed the subsequent live proxied HTTP action. The lower panel reports mean observable policy regret and live completion for all A/B/C conditions, highlighting the B-to-C structural-control comparison. Here, 0 ms regret denotes zero excess observable operational cost, not zero HTTP duration.

The counterbalanced replication also exposes the baseline pattern that verified state corrects. Across A and B, models selected `tool_alpha` in 17 of 18 decisions, including 7 of 8 cases in which `tool_beta` was listed first. This is an observed default route-label preference in this cohort, not a claim about universal positional bias. Condition C removed its operational consequence by selecting the healthy route in every regime regardless of which route was degraded or listed first.

5. Implications, boundaries, and agenda

Modern observability systems collect traces, metrics, and logs for human operators. Harness Awareness supplies the missing control-plane step: select the facts relevant to the immediate decision, preserve provenance and freshness, and expose the bounded result to the agent without giving it authority over truth or enforcement. The interface can support alternate-region routing, deadline-aware planning, quota-aware work decomposition, context-budget checkpointing, resource-aware concurrency, and multi-agent backpressure.

The evidence is strong for the tested mechanism: current verified tool-health facts improve selection between functionally equivalent routes. The primary cohort uses a synthetic host-owned monitor; the replication uses a minimal local HTTP service and deterministic proxy faults. Neither study claims uncontrolled production-outage rates, universal service topologies, or completed evidence for hardware, runtime, and economic state. Those are next high-value tests, not prerequisites for the demonstrated control-loop effect.

6. Conclusion

Agent Harness Awareness closes an operational loop that tool-using systems often leave open: the harness observes execution state while the agent chooses as if that state did not exist. In the frozen controlled cohort, verified state substantially outperformed both the task-only deployment baseline and the neutral same-shape structural control; the latter shows that the improvement is attributable to verified operational information rather than interface formatting. In the separately reported OpenTelemetry/Toxiproxy replication, that same information intervention selected the observable-best route and completed every C action across three providers and three fault regimes. The evidence supports a clear systems direction: turn selected, verified telemetry into an active agent control input while keeping truth and enforcement in the harness.

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